DSAA 4011 · LAB 6 · FRIDAY 16 OCTOBER 2026

# Teaching a Model to Reason

## Way 2 of Lecture 6 inside a reasoning language model

A model is trained on questions and final answers only. Today: how it can still learn the **steps** between, in one tiny maze.

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**Builds on Lecture 6: When the E-step is too big**

# Outline


- **The problem: the steps are hidden**

The model: a walker in a maze

The method: walk, keep, count

What it learns, and where it fails

The same loop in today's models

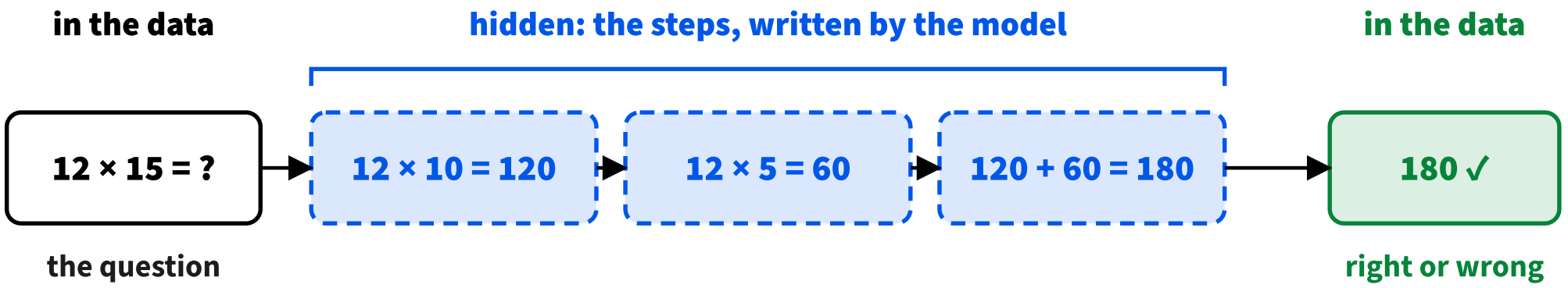

One question runs through the lab: how can a model learn steps it never sees?

# Main Question

**A reasoning model is trained on questions with answers only. Where does its reasoning come from?**

# 1 · The problem: the steps are hidden

**Trouble:** **the data shows the question and the answer, not the steps.**

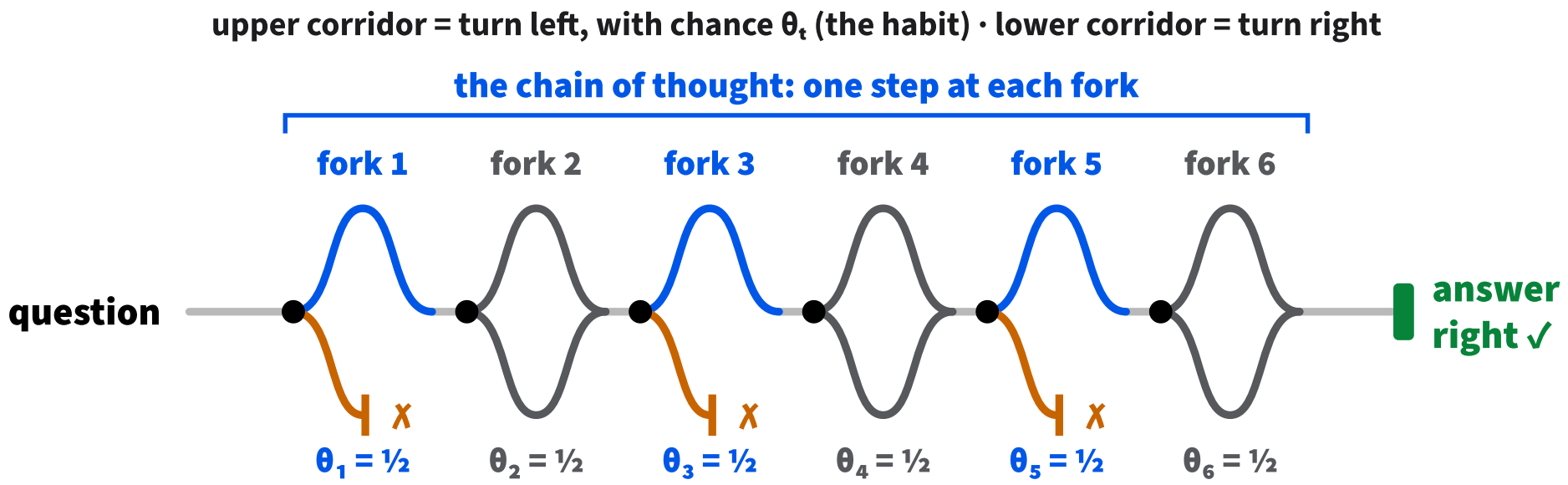



**a real chain of thought has hundreds of steps, and each could have been written another way**

**Look:** the steps play the part of Lecture 5's hidden coins, so to learn good steps we must first **guess** them, as expectation–maximization (EM) does.

# 2 · The model: a walker in a maze

## Each fork is one reasoning step; the exit is a right answer.




**Look:** the six habits $\theta_1, \ldots, \theta_6$ are the model's **weights**. Our goal: habits that make the exit likely.

# 3 · The method: what EM needs

## Trouble: the guess step weighs every path, and there are $2^L$.

**GUESS STEP** for every path z, its chance given the exit, p(z | exit).

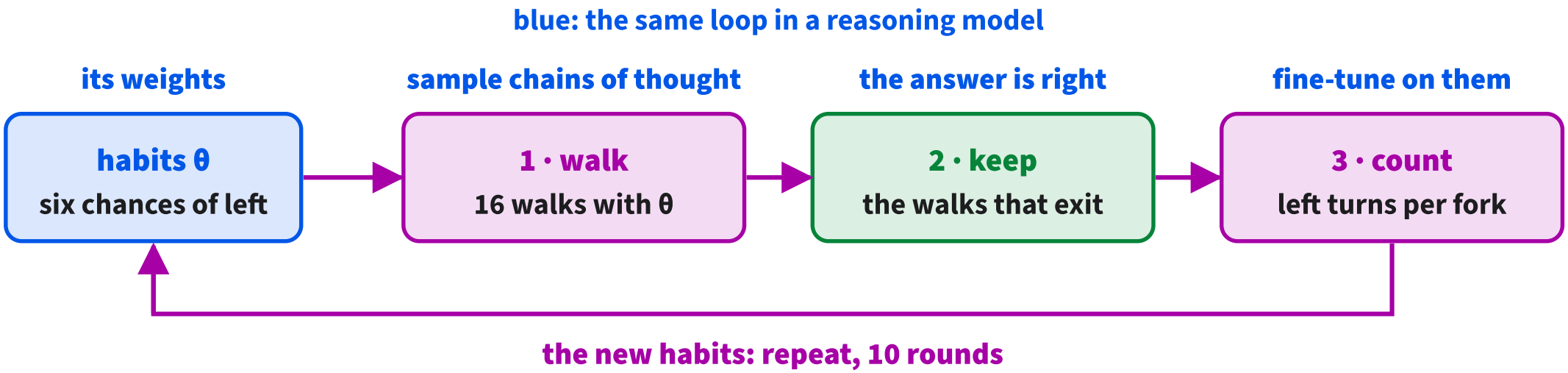
**COUNT STEP** count the turns at each fork, each path weighted by that chance.

| steps L | paths $2^L$ | |
|---|---|---|
| 6 (our maze) | 64 | a computer lists them at once |
| 20 | about a million | a short chain of thought |
| 30 | about a billion | a long chain of thought |

**Read:** two choices at every step double the list each time, so for a real chain of thought the guess step cannot be written out. This is expectation–maximization (EM) [Dempster, Laird and Rubin, 1977].

# 3 · The method: walk, keep, count

## Fix (Way 2): walk, keep the walks that exit, count the turns.




**count, one imaginary walk each way: $\theta_t \leftarrow$ (left turns at fork t among kept + 1) / (kept + 2)**

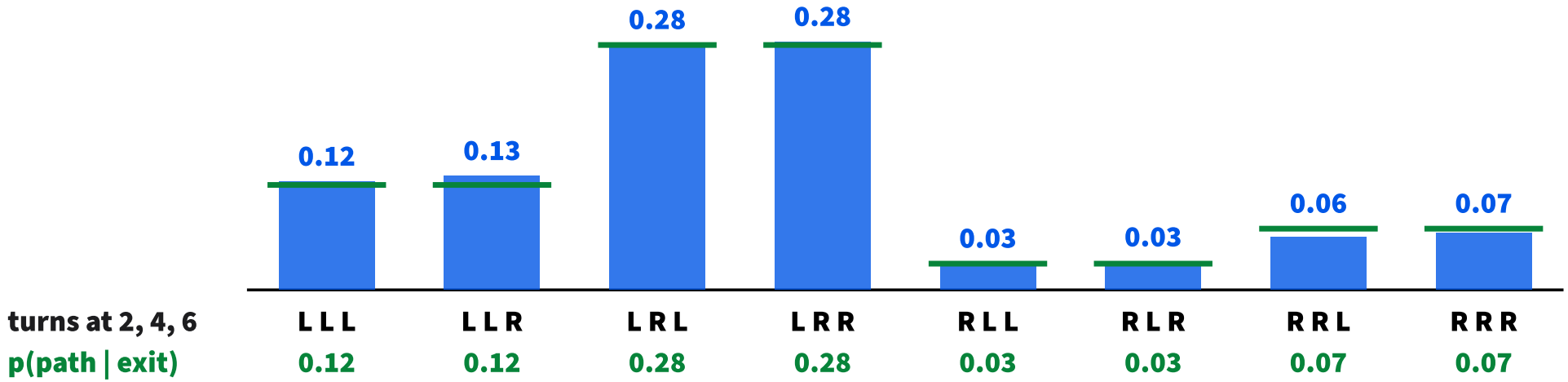
**Read:** no list of paths: we only run the walker forward and look at the exit. This is Monte Carlo expectation–maximization (EM) [Wei and Tanner, 1990].

# 3 · The method: why the count is right

## The kept walks follow p(path | exit), the guess step's chances.

$$\Pr(\text{a kept walk is path } z) = \frac{p(z) \cdot \mathbf{1}[z \text{ exits}]}{\Pr(\text{exit})} = p(z \mid \text{exit})$$

**habits: forks 1, 3, 5 at ½ · $\theta_2$ = 0.8, $\theta_4$ = 0.3, $\theta_6$ = 0.5 · 16 000 walks, 2 028 kept**




**blue bar: share among the kept walks · green line: p(path | exit), computed from the habits**

**Read:** a path is walked with chance p(z) and kept only if it exits, so among the kept walks it appears with chance p(z | exit). Counting kept walks is therefore the count step of expectation–maximization (EM), plus sampling noise **(Lecture 6, p. 29)**.

# 4 · Round 1 by hand

## Sixteen walks with every habit at ½; then keep and count.

| walk | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| fork 1 | L | R | L | L | R | R | R | R | L | R | L | R | L | R | L | R |
| fork 2 | R | L | R | R | R | R | L | L | R | L | L | L | L | R | L | L |
| fork 3 | R | R | L | R | R | L | L | R | L | L | L | R | L | L | R | R |
| fork 4 | R | R | L | R | L | L | L | L | L | R | L | L | L | R | L | L |
| fork 5 | L | R | R | L | L | L | R | L | L | R | L | R | R | L | R | L |
| fork 6 | L | R | L | R | R | L | L | R | L | L | L | R | L | L | R | R |
| exit? | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | | ✗ | | ✗ | ✗ | ✗ | ✗ | ✗ |

**kept: 2 of 16 walks exit (green) · fork 1: left in 2 of 2 → $\theta_1 = (2 + 1) / (2 + 2) = 0.75$**

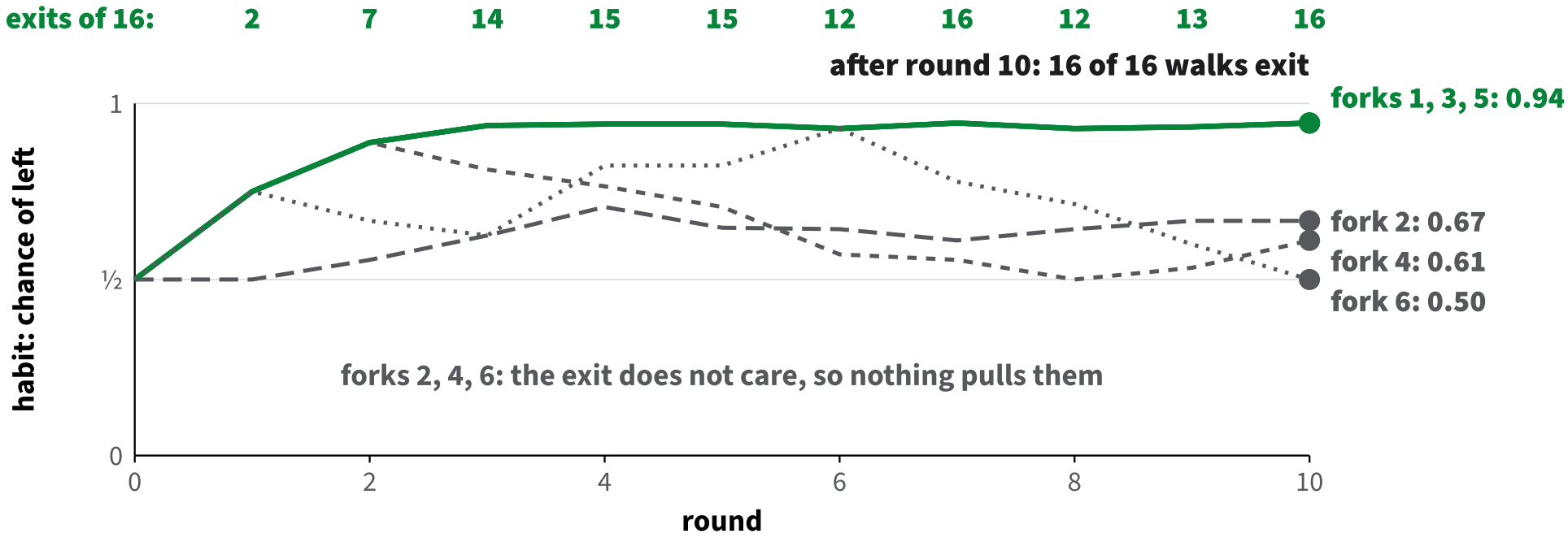
**fork 2: left in 1 of 2 → $\theta_2 = (1 + 1) / (2 + 2) = 0.50$ · the same count at every fork**

**Watch:** each column is one walk; an orange R at fork 1, 3 or 5 is a dead end; the green columns are kept. One imaginary walk each way keeps a habit off 0 and 1.

**Q: How many of the 16 walks exit? A: About 2: a walk needs left at three forks, ½ · ½ · ½ = 1 in 8, and 16 / 8 = 2.**

# 4 · Ten rounds

## Result: forks 1, 3, 5 learn to turn left; forks 2, 4, 6 do not.



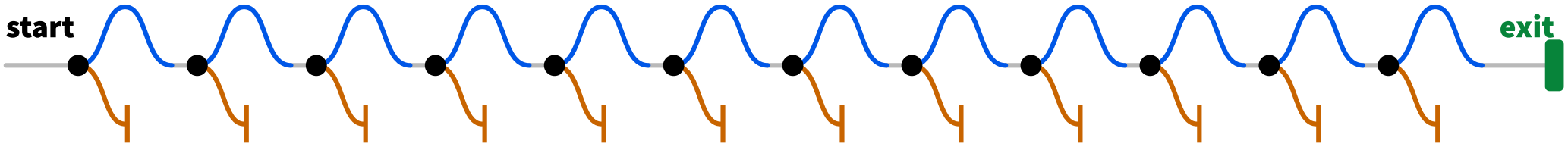

**Watch:** every kept walk turned left at forks 1, 3, 5, so their three habits rise together (one green line), and the walks that exit rise from 2 to 16 of 16. Exact expectation–maximization (EM) ends at 0.93 and ½ **(p. 21)**.

**Q: Why do forks 2, 4, 6 not learn? A: Both turns there reach the exit, so the answer check cannot tell them apart: they only drift with the sampling noise.**

# 4 · Where it fails: a cold start

**Trouble:** twelve forks, no walk exits, nothing to count.

**12 forks that all matter: the exit needs a left turn at every one**



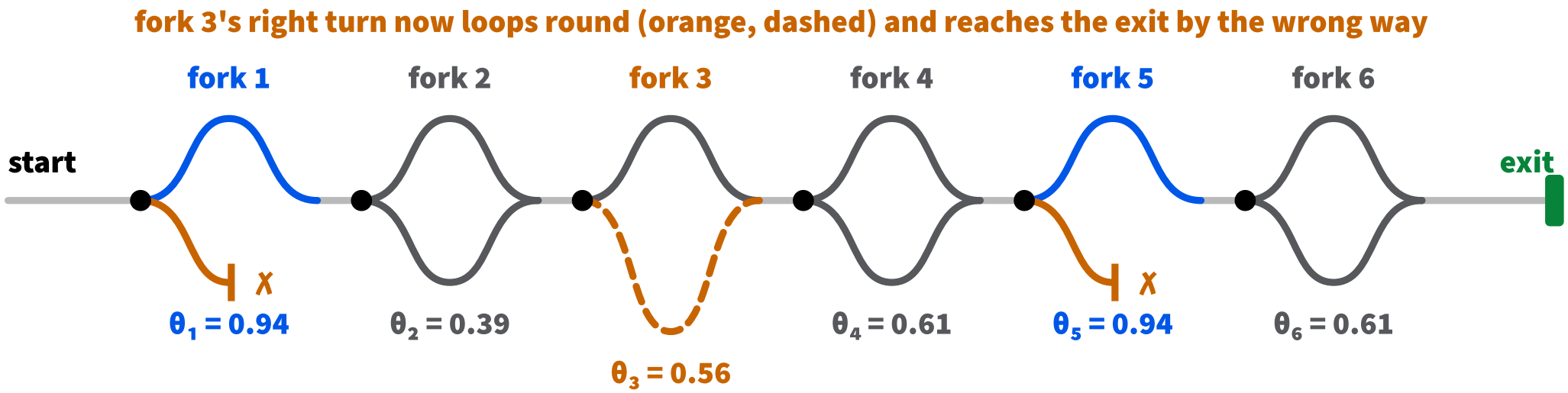

**16 walks a round, 10 rounds: kept 0 0 0 0 0 0 0 0 0 0 of 16: nothing to count, the loop stalls**

**4 096 walks a round: kept 1, 31, 2823 in rounds 1, 2, 3: one lucky walk, then the loop moves**

**Read:** at habits ½ one walk in $2^{12} = 4\,096$ exits, so 16 walks keep nothing and the habits cannot move **(Lecture 6, p. 57)**. The cure: easier problems first.

# 4 · Where it fails: a shortcut

**Trouble:** **wrong reasoning, right answer, kept and learned.**




round 10: 16 of 16 walks kept, 7 of them through the loop; after 10 rounds $\theta_3 = 0.56$, not near 1

**Read:** the answer check keeps every walk that exits, so walks through the loop are learned too.

**Q: How would you catch it? A: Check the steps too: keep a walk only if it turned left at fork 3; then $\theta_3 = 0.94$. This is process supervision [Lightman et al., 2023].**

# 5 · The same loop in today's models

## Result: the same loop trains today's reasoning models.

| system | walk | keep | count |
|---|---|---|---|
| STaR: Self-Taught Reasoner | write one chain of thought per question | the chains whose answer is right | fine-tune on them; repeat |
| ReST-EM: Reinforced Self-Training | sample many chains per question | the chains that pass a check | fine-tune the base model; a few rounds |
| Llama 3 | sample several answers per prompt | the best, by a learned judge | fine-tune; next round |
| DeepSeek-R1 | sample chains from a trained model | the correct, readable ones | fine-tune, then more reinforcement learning |

**Read:** the maze's two lessons carry over: easy questions first against the cold start, a check of the steps against the shortcut.

This is STaR [Zelikman et al., 2022] and ReST-EM [Singh et al., 2024]; Llama 3 [Grattafiori et al., 2024] and DeepSeek-R1 [DeepSeek-AI, 2025] run a rejection-sampling stage.

# Conclusion

**TROUBLE** The steps are **hidden**, and there are too many paths for the guess step of expectation–maximization (EM).

**FIX** Way 2: **walk, keep, count**. The kept walks follow p(path | exit), so counting them is the count step of EM.

**RESULT** The steps that decide the answer are learned; a cold start stalls; a shortcut is learned too, so **check the steps**.

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**Remember:** a model learns only what the answer check can see.

# Your turn

## Predict, then run the maze.

**Experiment 1: the maze walker** (`python lab6_explore.py 1`, course site → Week 6 → Lab files)

- 64 walks a round instead of 16: do the free forks wander less?
- Fork 1 starts at habit 0.1: how many rounds until it is found?

**Other problems**, experiments 2 to 6

2. A noisy picture: Ways 2, 3 and 4.
3. The bits in your phone: messages.
4. Reading a library: guess alone.
5. Ranking players: one match at a time.
6. Messages that swing: damping.

**Tip:** write your guess first; surprises teach.

# Selected References


DeepSeek-AI (2025).
DeepSeek-R1: incentivizing reasoning capability in large language models via reinforcement learning.
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Maximum likelihood from incomplete data via the EM algorithm.
*Journal of the Royal Statistical Society, Series B (Methodological), 39(1):1–38.*

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The Llama 3 herd of models.
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Let's verify step by step.
*arXiv:2305.20050.*

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Beyond human data: scaling self-training for problem-solving with language models.
*Transactions on Machine Learning Research.*

Wei, G. C. G. and Tanner, M. A. (1990).
A Monte Carlo implementation of the EM algorithm and the poor man's data augmentation algorithms.
*Journal of the American Statistical Association, 85(411):699–704.*

Zelikman, E., Wu, Y., Mu, J., and Goodman, N. D. (2022).
STaR: bootstrapping reasoning with reasoning.
*In Advances in Neural Information Processing Systems 35.*

# Thank you!

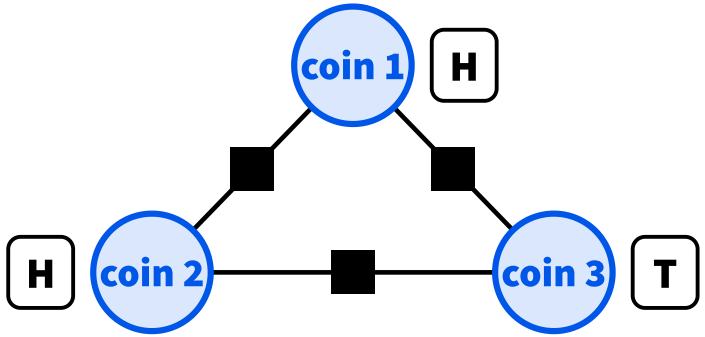
Next: Lecture 7. Way 2 returns whenever a model can only be run forward

Backup: the ring, which way can run, the count by hand, the imaginary walks, exact against sampled, other problems.

# Backup: Lecture 6's ring

## Eight stories were listed; Way 2 is the one that needs no list.

**hidden: three coins, A or B**
**seen: the flips H, H, T**
**links: keep ¾ on every link**




**the question: coin 1's chance of being A**




**Look:** Way 2 replayed the game, kept the runs that showed H, H, T and counted **(Lecture 6, p. 29)**; it is noisy but right on average.

# Backup: which way can run?

## Only Way 2 runs when the only tool is the model.

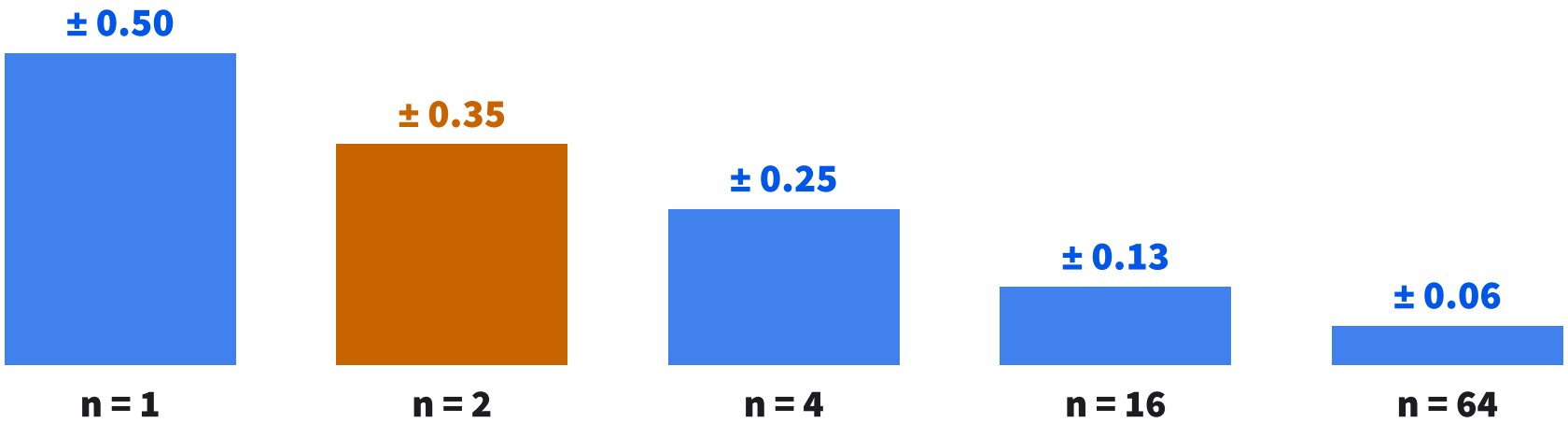
| way | what it needs | in a reasoning model? |
|---|---|---|
| 1 · messages along a line | a line: each step touches only the step before | **no: there is no line** |
| 2 · act it out | running the model forward, and the answer check | **yes: run it forward** |
| 3 · everyone guesses alone | each step touching only a few others | **no: each touches all** |
| 4 · messages in a circle | small links between a few steps | **no: each touches all** |

**Read:** a language model writes the next step from everything written so far, so every step touches all earlier ones: no line for Way 1 **(Lecture 6, p. 27)**, no small links for Ways 3 and 4.

# Backup: why the imaginary walks

## Two kept walks make a noisy count: add one imaginary walk each way.

**typical error of a counted chance (truth ½) from n kept walks: √(½ · ½ / n)**




**round 1, fork 6: both kept walks turned left: raw count 2 / 2 = 1.00, as if fork 6 were sure**

**with one imaginary walk each way: $\theta_6 = (2 + 1) / (2 + 2) = 0.75$, held back from certainty**

**Read:** by Theorem 7 **(Lecture 6, p. 31)** the error falls like $1/\sqrt{n}$, so two walks say little; and a habit of exactly 1 could never change again.

# Backup: exact EM against sampling

## Exact expectation–maximization (EM) over all 64 paths agrees where it matters.

| | exact EM, all 64 paths | | walk, keep, count | |
|---|---|---|---|---|
| **fork** | **round 1** | **round 10** | **round 1** | **round 10** |
| 1, 3, 5 (matter) | 0.75 | **0.93** | 0.75 | **0.94** |
| 2 (free) | 0.50 | 0.50 | 0.50 | 0.67 |
| 4 (free) | 0.50 | 0.50 | 0.75 | 0.61 |
| 6 (free) | 0.50 | 0.50 | 0.75 | 0.50 |

**Read:** exact EM uses the same count, with the kept walks replaced by their average over all 64 paths: the forks that matter agree, and the free forks stay at ½ while sampling wanders, the noise of Theorem 7 **(Lecture 6, p. 31)**.

# Backup: the count step of round 1 by hand

## Round 1: two kept walks, L R L L L L and L L L L L L.

$$\theta_t \leftarrow \frac{\text{left turns at fork } t \text{ among the kept walks} + 1}{\text{kept walks} + 2}$$

> the count step (Lecture 6, p. 30)

$$\theta_1 = \frac{2+1}{2+2} = 0.75, \qquad \theta_2 = \frac{1+1}{2+2} = 0.50$$

> fork 1: both turned left; fork 2: one did

$$\theta_3 = \theta_4 = \theta_5 = \theta_6 = \frac{2+1}{2+2} = 0.75$$

> both turned left at forks 3 to 6

$$\text{without the imaginary walks: } \theta_1 = \frac{2}{2} = 1.00$$

> a habit at 1 could never move again

**Notice:** the exact guess step gives 1 at forks 1, 3, 5 and ½ at 2, 4, 6; two walks cannot tell fork 3 from fork 4.

# Backup: Theorem 7's error for n kept walks

## A counted chance p from n kept walks is off by about √(p(1 − p)/n).

| kept walks n | p = ½: typical error | p = 0.9: typical error | where it happens in the deck |
|---|---|---|---|
| 1 | ± 0.50 | ± 0.30 | the cold start with 4 096 walks, round 1 |
| 2 | ± 0.35 | ± 0.21 | round 1 (p. 9) |
| 4 | ± 0.25 | ± 0.15 | the shortcut maze, round 1 (p. 12) |
| 16 | ± 0.13 | ± 0.08 | rounds 7 and 10 (p. 10) |
| 64 | ± 0.06 | ± 0.04 | the kit, 64 walks a round |

**Notice:** four times the kept walks halve the error (Lecture 6, p. 31); the free forks wander by about this much every round, which is the wandering seen over ten rounds.

# Backup: five other troubles, in the kit

## Five other troubles, in the kit as experiments 2 to 6.

| | Trouble | Fix | Where |
|---|---|---|---|
| 2 | a noisy picture | redraw, guess alone or messages (Ways 2–4) | photos, scans |
| 3 | the bits in your phone | messages in a circle (Way 4) | phones, Wi-Fi, space |
| 4 | reading a million documents | guess alone, then count (Way 3, then count) | search |
| 5 | ranking players from wins | the truth judges one match at a time | matchmaking |
| 6 | messages that swing | damping: half old, half new | any cycles |

**Read:** each is one of Lecture 6's four ways on a deployed system; the kit's README states a prediction for each, and the answers after it.